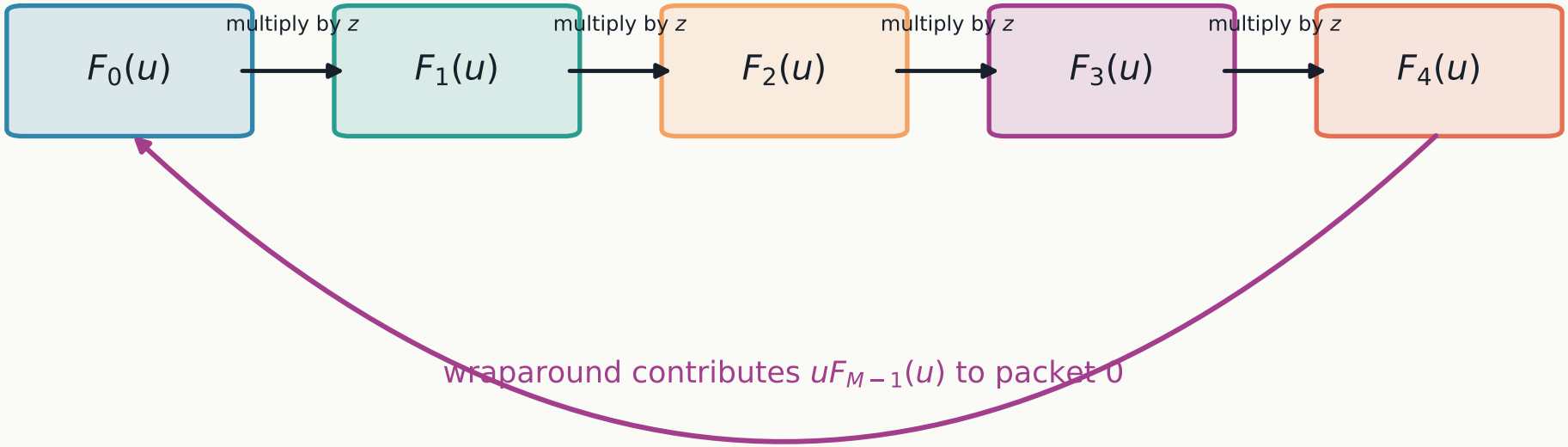The carry-shift matrix $C_M(u)$ is multiplication by z in packet coordinates